UPI-NETRA

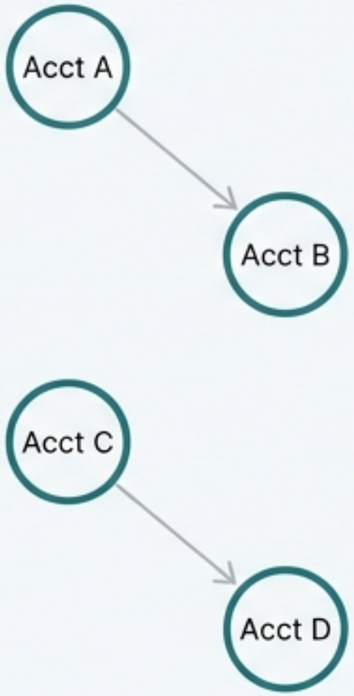
Real-Time Intelligence for Mule & Collusive Fraud

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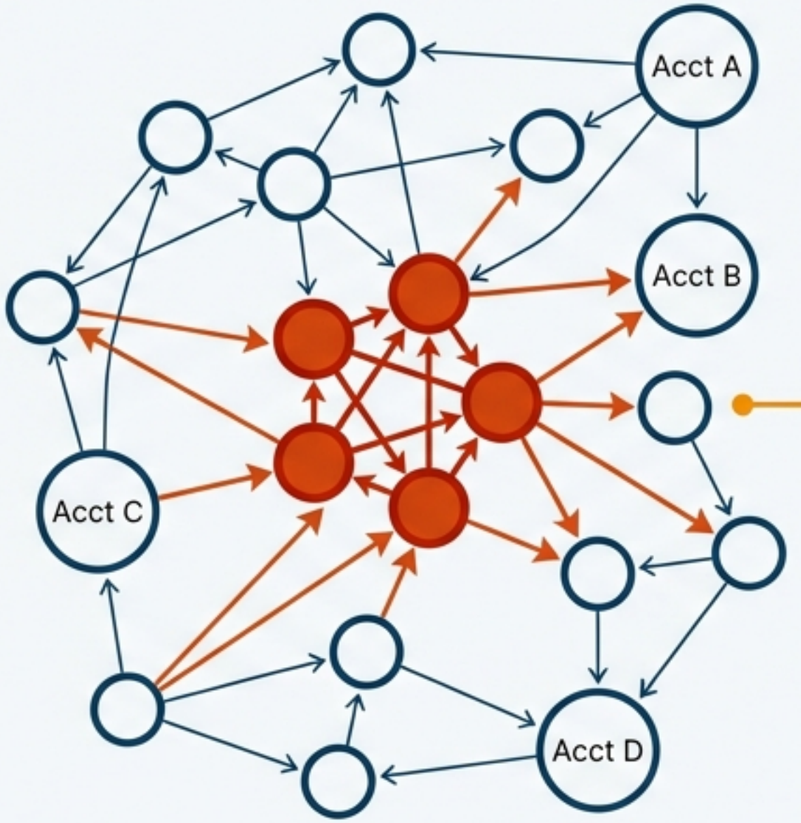
Today’s Fraud Systems See Transactions, Not Collusion

Current systems are reactive, analyzing accounts in isolation. They are fundamentally blind to coordinated, network-based attacks that emerge across multiple accounts.

The Old Way: Account-Level View



The Hidden Reality: Network-Level Threat

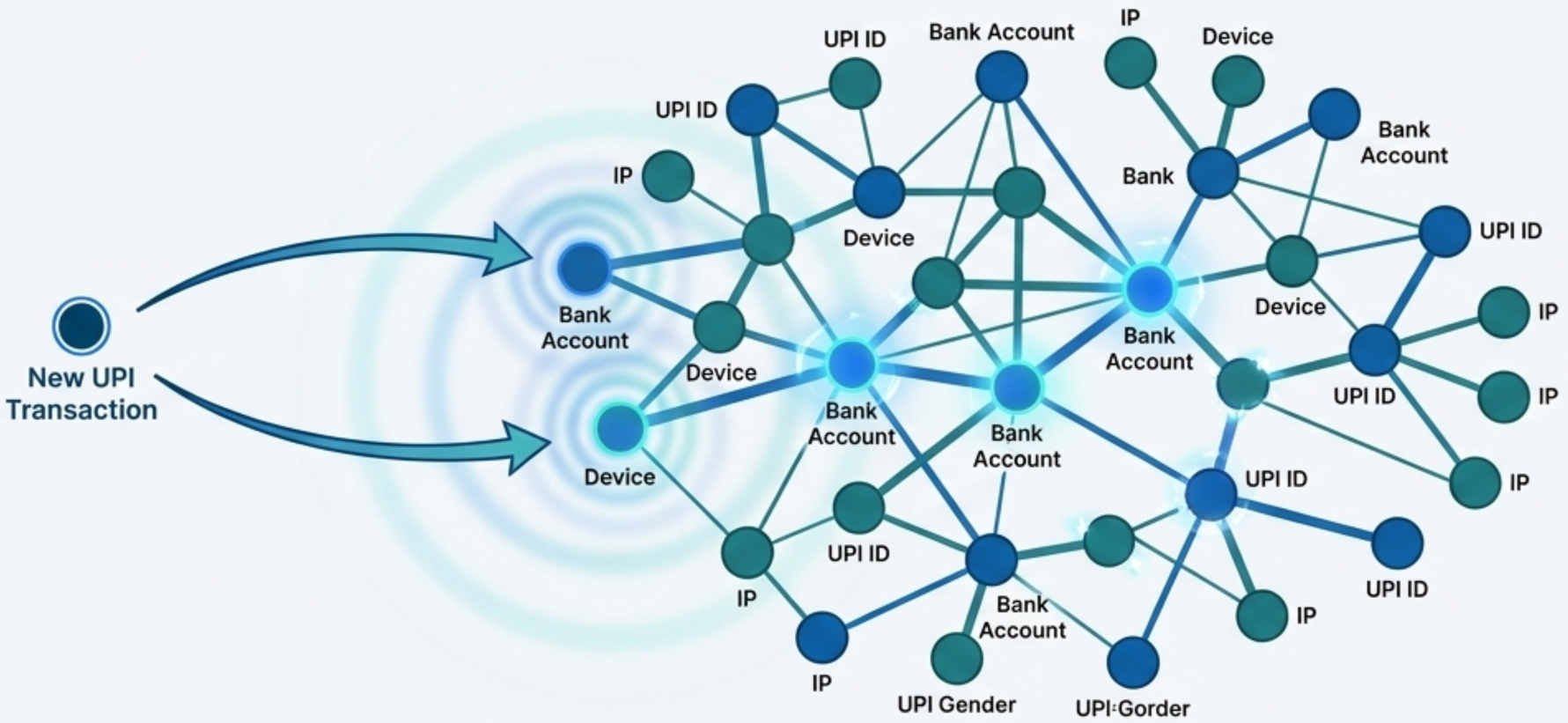


They miss fraud as it's forming, only catching it after the money is gone.

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Our Paradigm Shift: Detect Fraud as an Emerging Graph Phenomenon

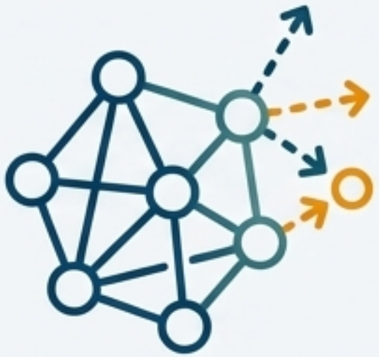
We are not building another fraud model. We are building **Fraud Intelligence Infrastructure**. Instead of batch analysis of past events, we build a **Living Transaction Graph (LTG)** that evolves in real-time with every UPI transaction.



This allows us to detect fraud ***while the graph is still forming***, not after losses are incurred.

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Four Pillars of Proactive Fraud Intelligence



Living Transaction Graph

A streaming graph engine that updates with every transaction, modeling the entire ecosystem in real-time.



Mule Account “Birth” Detection

Identifies mule accounts at the moment of creation by analyzing how they connect to the existing graph, not after patterns stabilize.



Collusive Ring Intelligence

Uncovers fraud rings by detecting specific graph patterns (“motifs”) and synchronized transaction timing, revealing coordinated behavior.



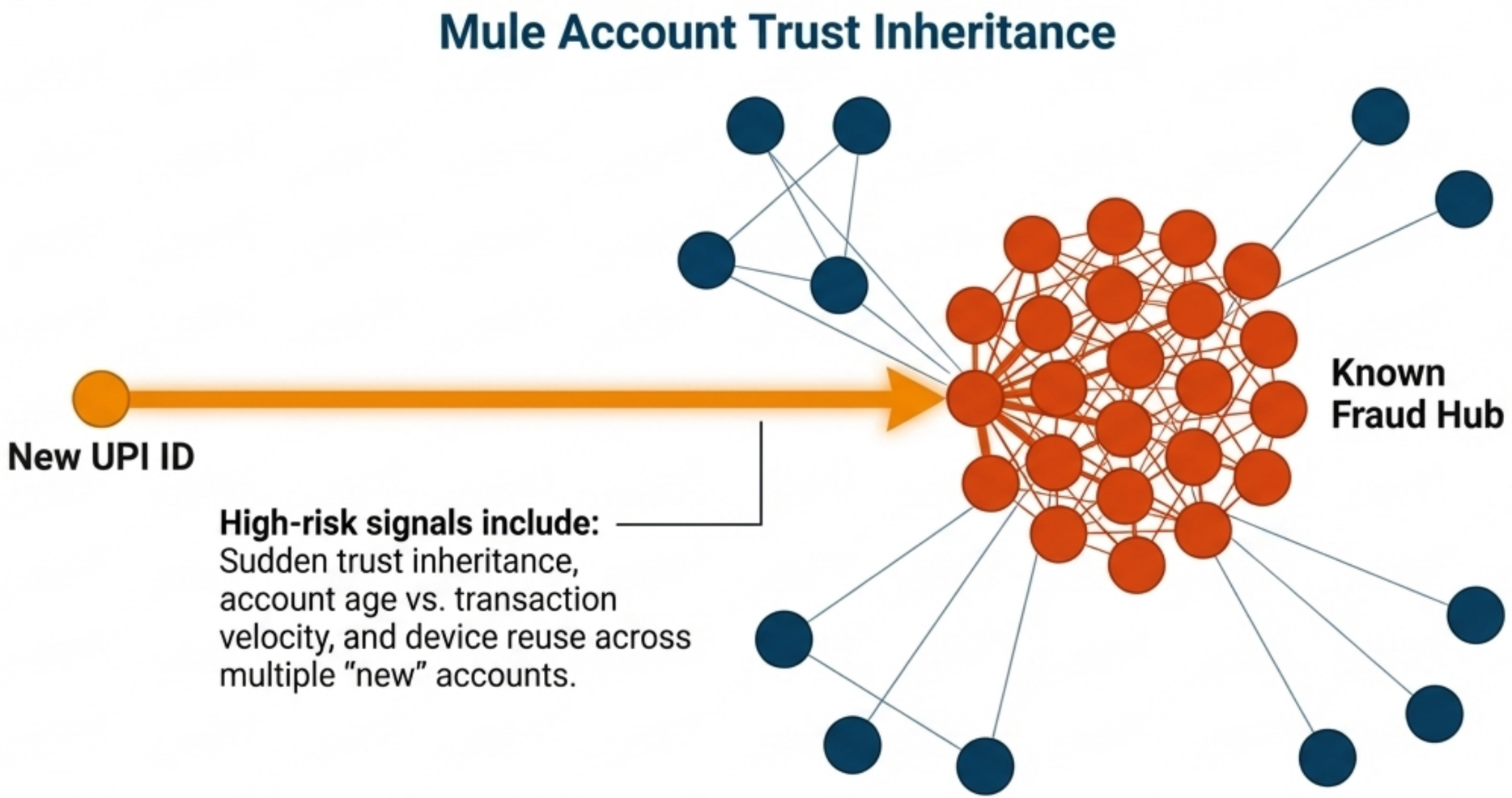
Device-Identity Shadow Graph

A privacy-preserving graph that roots identity in the device, exposing mule farms and device-sharing schemes.

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We Detect Mule Accounts at Birth, Not After Damage is Done

Legitimate users build trust and network connections slowly. Mule accounts attach directly to existing fraud hubs.



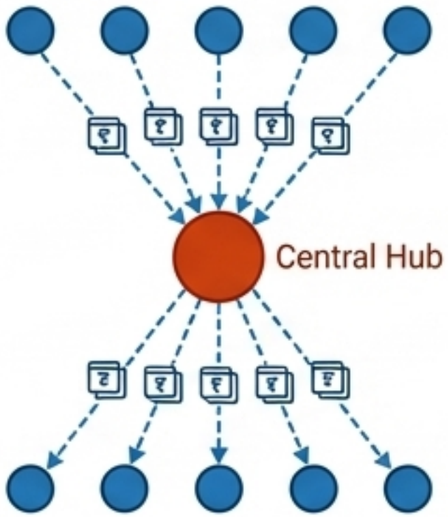
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Collusion is Synchronized Behavior, Not Just Money Flow

We combine graph structure analysis with temporal analysis to find behavior that is too coordinated to be human.

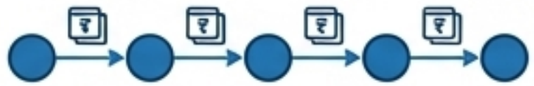
Fraud Motifs

Fan-in → Fan-out Rings



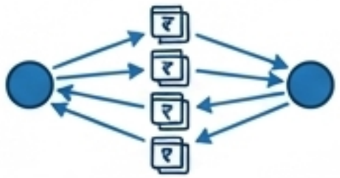
A central node acts as a conduit, receiving funds from multiple sources before quickly distributing them to numerous destinations.

Daisy Chains



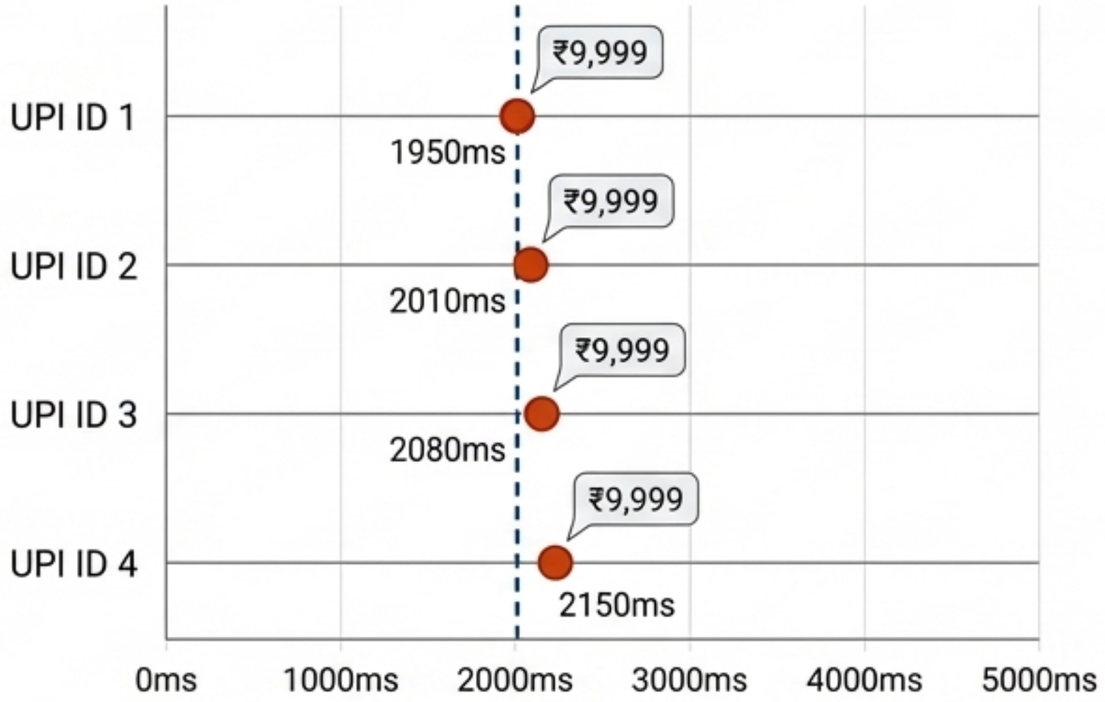
Funds are moved sequentially through a series of accounts in a linear fashion, obscuring the original source.

Burst-based "Wash" Flows



Two or more accounts engage in a rapid series of small, reciprocal transactions to inflate transaction volume or hide the nature of funds.

Temporal Collusion

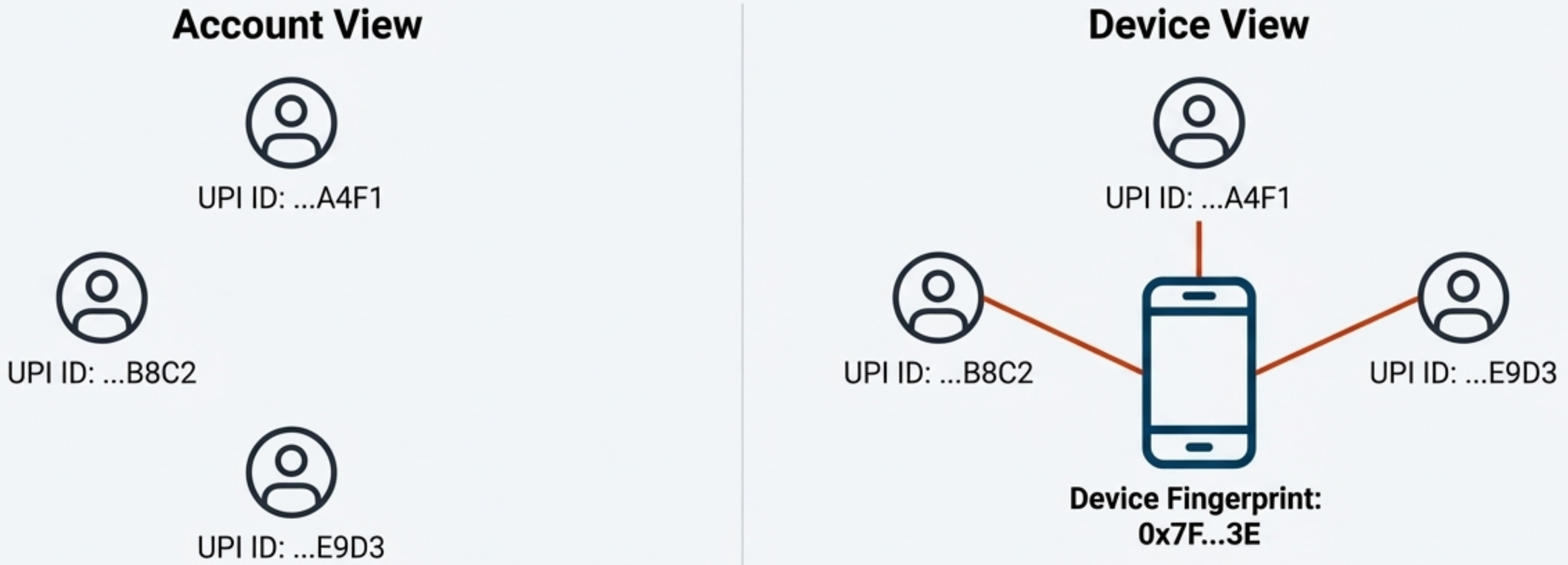


Legit users are random.
Fraud rings are clock-driven.

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The Device is the True Root of the Fraud Graph, Not the Account

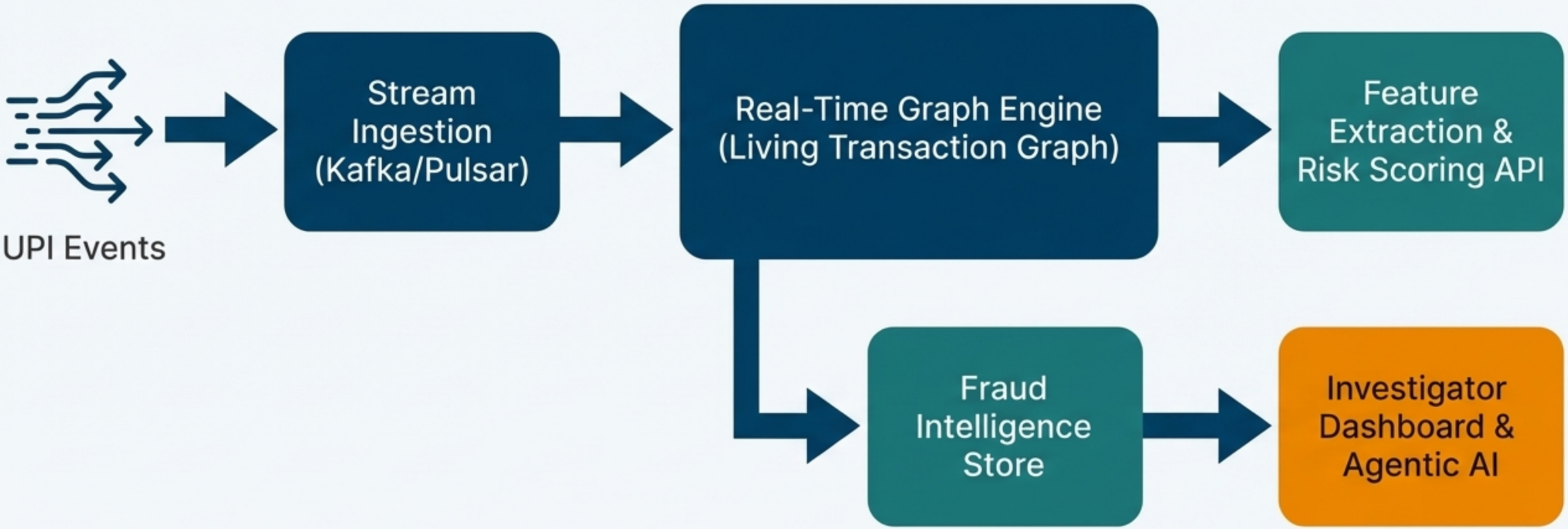
Key Innovation: We create a parallel **Device-Identity Shadow Graph**, shifting the focus from accounts to the hardware they run on.



All identifiers (Device Fingerprint, SIM Hash, IP) are cryptographically hashed and tokenized.
The system is privacy-preserving and designed to be RBI-compliant.

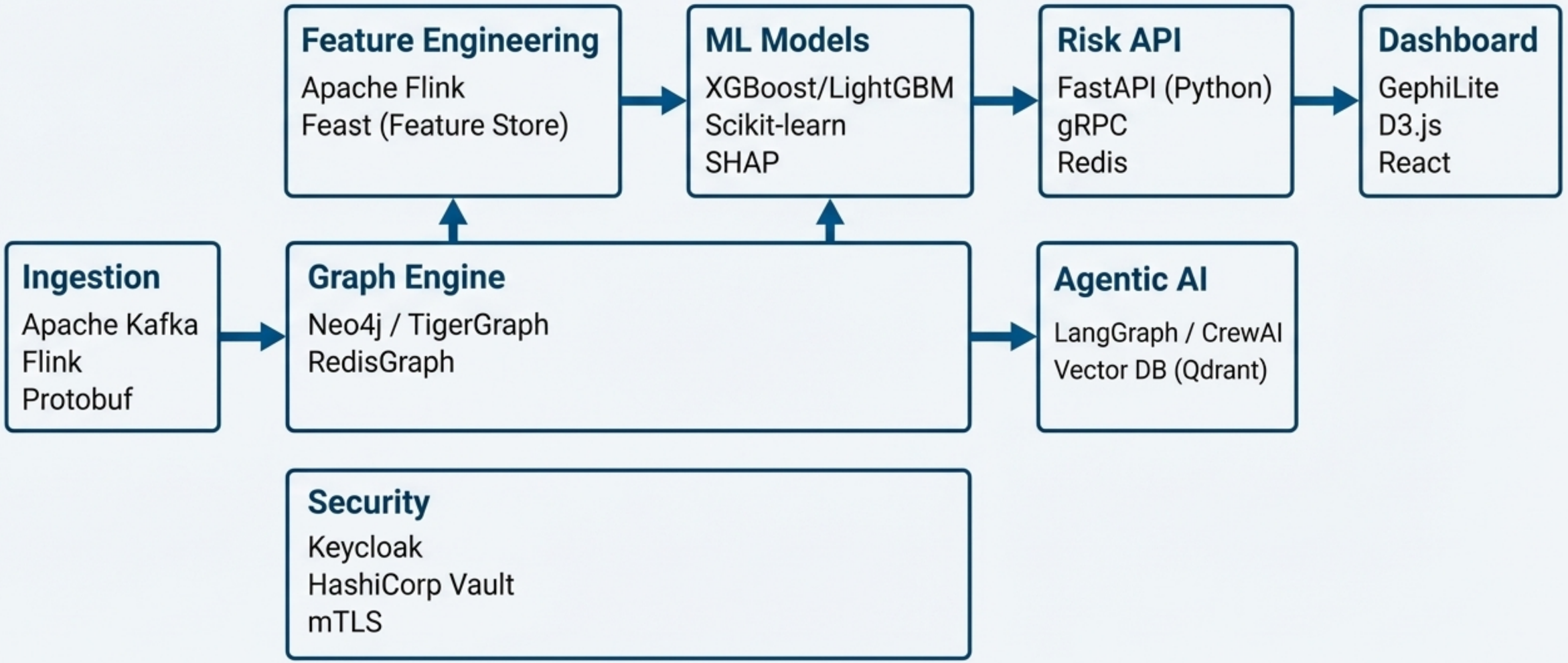
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Architecture Built for Real-Time Intelligence at India-Scale



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A Technical Stack Built on Proven, Scalable, and Open Technologies



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A Single API Call Delivers an Actionable Risk Score in Under 40ms

POST /upi/risk-score

```
{
  "payer_id": "user1@bank",
  "payee_id": "merchant5@bank",
  "amount": 4999.00,
  "device_hash": "0x8A2...F9B",
  "timestamp": "2023-10-27T10:00:00Z"
}
```

200 OK Response

```
{
  "upi_risk": 0.94,
  "ring_probability": 0.88,
  "mule_stage": "ACTIVE_MULE",
  "reason_codes": [
    "SHARED_DEVICE_WITH_FLAGGED_NODES",
    "TEMPORAL_COLLUSION_DETECTED"
  ],
  "recommended_action": "DELAY_AND_REVERIFY"
}
```

<40ms Response Time

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The Agentic AI Investigator: Your Autonomous Fraud Analyst

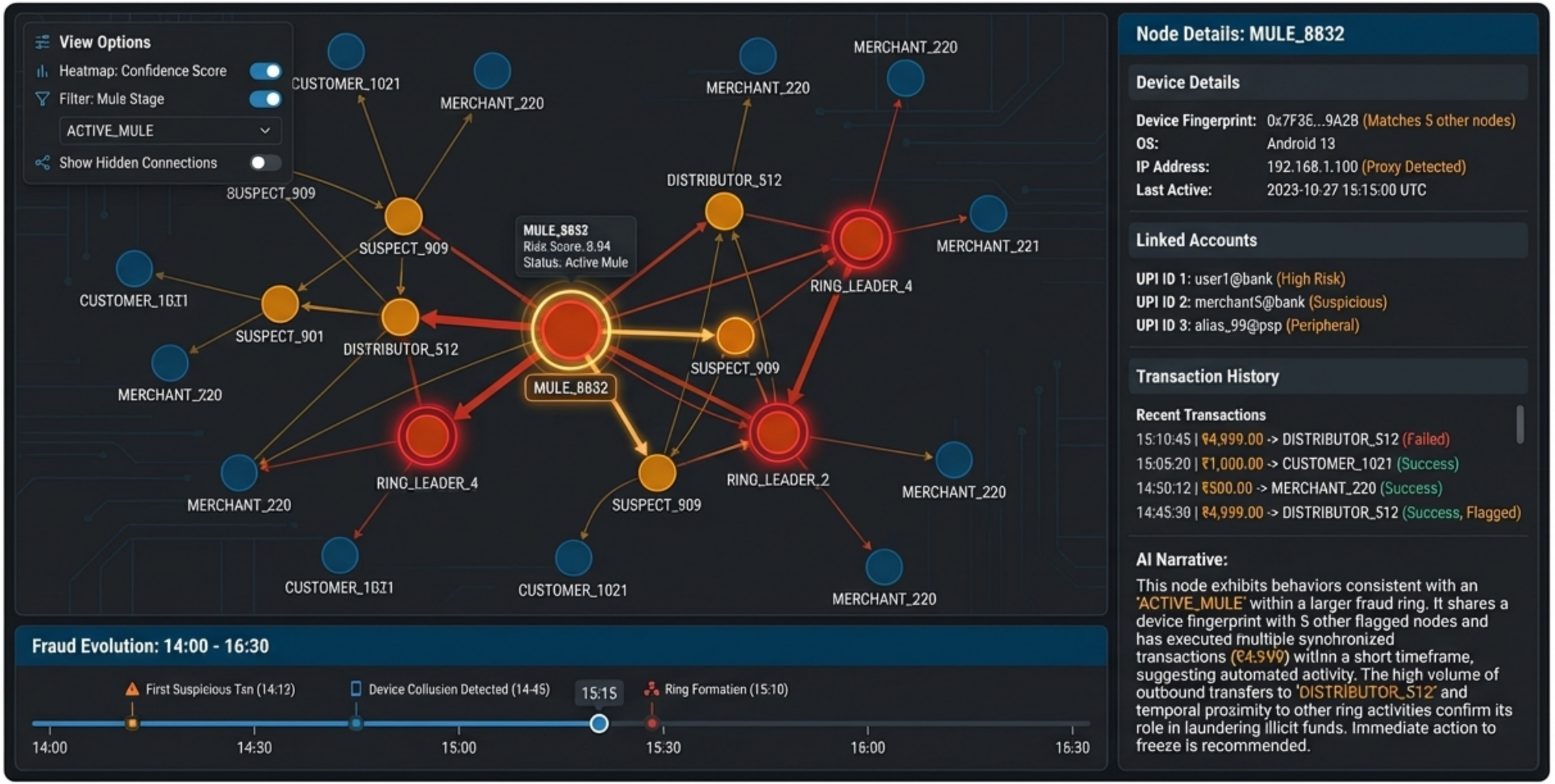
A multi-agent AI system automates the initial investigation, connecting dots across the graph, behavior, and time to produce human-readable intelligence intelligence.



> This UPI ID is part of a **17-node ring** sharing 2 devices, executing **synchronized ₹4,999 transfers every 90 seconds**. We classify this as an '**Active Mule**' in a '**Distributor**' role.

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The Investigator Dashboard: Visualizing Fraud as it Evolves



Node Details: MULE_8832

Device Details

Device Fingerprint: 0x7F3E...9A2B (Matches 5 other nodes)
OS: Android 13
IP Address: 192.168.1.100 (Proxy Detected)
Last Active: 2023-10-27 15:15:00 UTC

Linked Accounts

UPI ID 1: user1@bank (High Risk)
UPI ID 2: merchant5@bank (Suspicious)
UPI ID 3: alias_99@psp (Peripheral)

Transaction History

Recent Transactions

15:10:45 | ₹4,999.00 -> DISTRIBUTOR_512 (Failed)
15:05:20 | ₹1,000.00 -> CUSTOMER_1021 (Success)
14:50:12 | ₹500.00 -> MERCHANT_220 (Success)
14:45:30 | ₹4,999.00 -> DISTRIBUTOR_512 (Success, Flagged)

AI Narrative:

This node exhibits behaviors consistent with an 'ACTIVE_MULE' within a larger fraud ring. It shares a device fingerprint with 5 other flagged nodes and has executed multiple synchronized transactions (₹4,999) within a short timeframe, suggesting automated activity. The high volume of outbound transfers to 'DISTRIBUTOR_512' and temporal proximity to other ring activities confirm its role in laundering illicit funds. Immediate action to freeze is recommended.

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Measuring What Matters: From Latency to Early Detection

Performance

- Detection Latency: <40ms
- Graph Expansion Speed: Nodes/sec

The Innovation Metric

‘Fraud Ring Discovery Time’

Time taken to detect a collusive network from the ***first fraudulent transaction***. This is our key differentiator.

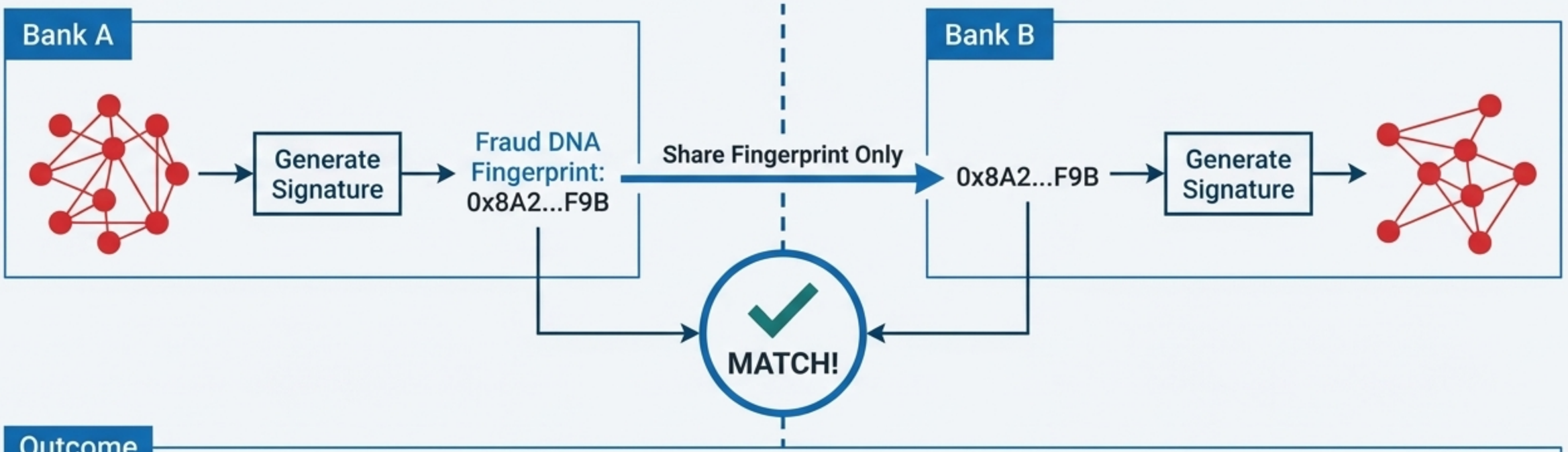
Effectiveness

- Precision / Recall vs. Baseline
- False Positive Reduction Rate

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Fraud DNA: Sharing Intelligence Across Banks Without Sharing Data

Each detected fraud ring is assigned a unique "**DNA Fingerprint**" — a mathematical signature based on its behavioral patterns (timing, amounts, graph density).



Outcome

This reveals a cross-bank attack campaign without either bank ever exposing customer data.

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UPI-NETRA: The Future UPI Security Layer

“UPI-NETRA is a real-time fraud intelligence platform that detects mule accounts and collusive UPI fraud rings by watching how networks, devices, and behaviors evolve — not just transactions.”

- **Proactive:** Detects fraud before loss.
- **Intelligent:** Uncovers collusion, not just alerts.
- **Explainable:** Delivers investigator-ready narratives.
- **Ready**: Built for India-scale and real-time integration.

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